Indian Institute of Technology, Bhilai

CSP203: Software Tools & Technologies Laboratory

Lab Activity - 9

Full Marks: 20 *Deadline: 05:30 PM, 11-09-2025*

Submission Instructions:

1) Answer all the questions in the lab activity. You have to prepare two latex files, one to answer questions of part 1 [10 Marks (2+2+2+2+2)] and another one to replicate part 2 [10 Marks (2+2+2+2+2)].
2) Write all the LATEX source files and name them as: <YourRollNo>_<LabActivityNo>_ <QuestionNo>.tex
3) For each question: compile the LATEX file and include both the source code and the generated PDF.
4) Compress (zip) all LATEX source files along with the .tex and bibliography files and output PDF into a single folder.
5) Name the zip file strictly as: <RollNo>_<Name>_ <ActivityNo>.zip
   Example: B24CS055_Shareen 04.zip
6) **Plagiarism is strictly prohibited.**

Objectives:

1. Gain proficiency in creating well-structured LATEX documents.
2. Learn using mathematical environments for equations, fractions, symbols.
3. Practice defining custom commands and macros for reusability.
4. Packages such as geometry, enumitem, and hyperref to enhance formatting.
5. Work with tables, lists, and section formatting to produce prof. reports.
6. Develop problem-solving skills by preparing diagrams, formulas, and formatted text in LATEX.

Part 1

Q1. Title, Abstract, and Structure

Create a LaTeX document with:

- A title block (title: "**Introduction to LaTeX**", your name as author, and automatic date).
- An abstract reflecting on the role of LaTeX in research writing.
- Three main sections: **Introduction**, **Methods**, and **Conclusion**, where **Methods** has two subsections: **Data Collection** and **Analysis**.

Q2. Cross-References with Equations

In the Introduction, refer to a "key formula" by its equation number. In the Results section, typeset Euler's identity:

$$e^{i\pi} + 1 = 0$$

Label the equation and ensure that the earlier reference points to it correctly.

Q3. Table

Create a table comparing the marks of three students (Alice, Bob, Charlie) in three subjects (Math, Physics, CS). The table must have borders, headings, and horizontal lines separating rows.

Q4. Algorithm

Write pseudocode in LaTeX that classifies an integer as positive, negative, or zero using `if, else if, else` conditions. Format it using the `algorithm` environment.

Q5. Bibliography and Citations

Create a separate file named `lab2.bib` with at least two references . In your LaTeX file, write a short paragraph that cites each reference at least once. Use both `\citet{}` and `\citep{}` to demonstrate the difference. Finally, include the bibliography in your document using an appropriate style.

Part 2

Q. Write the Latex source code to generate the following pdf csp203_part2.pdf